

Daniel R. Russell

Founder-operator and applied ML/DL engineer building **SNPTX**, an MLOps-first orchestration framework for multi-modal biomedical ML - a domain where PII/PHI-adjacent handling, provenance-tracked data, and governed access are load-bearing. Deployment pathway ships as FastAPI inference with RBAC (HMAC-signed tokens) and a hash-chained audit log at 21 CFR Part 11 posture, lab-hosted via Terraform + Helm. MLOps spine: Snakemake DAGs, DVC content-hashed data, MLflow tracking, DuckDB experiment catalog, contract-validated extension runner, autonomous evaluation with SPRT stopping. Runs on AWS / GCP. Supported by the **Harvard Innovation Labs** while completing the Harvard ALM in Data Science & Artificial Intelligence. Previously founded **FIVSS**, acquired by **Booking Holdings**. US citizen.

Recent Role • Founder & Systems Engineer, SNPTX • ML/DL framework for biomedical discovery (GNNs, DL, LLM/NLP, Autonomous Learning System) • **Harvard Innovation Labs**

Technical Focus • PyTorch • GNNs • Deep Learning • NLP/LLM workflows • Snakemake/DVC/MLflow • AWS/GCP Parallel GPUs and GPU clusters

Postgraduate • Harvard ALM in Data Science & Artificial Intelligence (HES '27) • MITx MicroMasters, Statistics & Data Science ('24), MIT Institute for Data, Systems and Society (IDSS)

Prior founder exit • FIVSS → Booking Holdings

Email • dan@snptx.ai • drr508@g.harvard.edu

Web • [snptx-academic](#) • [github @snptx1](#)



At a glance

A quick read on the work: MLOps-first pipelines for biomedical ML with provenance-tracked data and hash-chained audit (21 CFR Part 11 posture), FastAPI + RBAC + Terraform / Helm deployment surface, LLM-agent tooling, and a decade of full-stack operator credibility from FIVSS through acquisition.

ACADEMIC SIGNAL

Harvard + MITx

Harvard ALM Data Science & AI, **4.00 GPA**; MITx MicroMasters, Statistics & Data Science.

COMPANY BUILDING

SNPTX + FIVSS

Active biomedical ML systems company; prior founder exit to Booking Holdings.

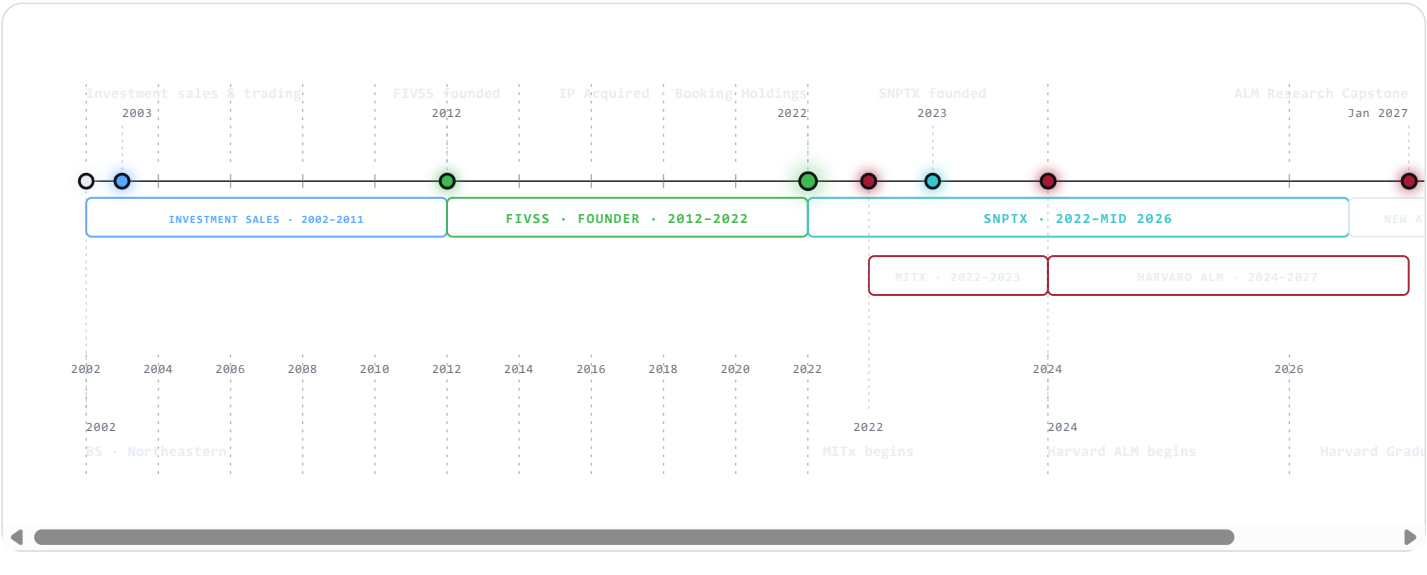
TECHNICAL DEPTH

GNNs + DAGs

PyTorch, biomedical graph ML, Snakemake/DVC/MLflow provenance, reproducible pipelines.

Career arc

A compressed early-career context followed by a deliberately recent-weighted founder, postgraduate, and applied-ML systems arc. The axis is intentionally non-linear: 2002–2022 occupies the left half, while the recent founder, postgraduate, and applied-ML period receives the full right half.



Professional experience

Reverse chronological. Founder-operator roles with full ownership of product, technology, and commercial workstreams.

SNPTX · Founder & Systems Engineer

HARVARD INNOVATION LAB

2022 – PRESENT

SNPTX · deterministic ML orchestration for biomedical discovery · supported by the [Harvard Innovation Labs](#) · Cambridge, MA

Founded and single-handedly built SNPTX, an MLOps-first orchestration layer engineered for regulated or governance-heavy environments: DVC content-hashed inputs (provenance-tracked), Snakemake DAGs with deterministic replay, MLflow experiment tracking, DuckDB experiment catalog, a contract-validated extension runner, an autonomous evaluation loop with SPRT stopping, and a deployment surface with RBAC (HMAC-signed tokens), hash-chained audit at 21 CFR Part 11 posture, and Terraform + Helm lab-hosted delivery. The four operational surfaces sit under a shared governance model; nothing on the public site is a clinical-performance claim.

- ▶ Architected a secure MLOps stack solo: DVC content-hashed data (leakage-free splits, lineage-stamped checkpoints), Snakemake DAGs, MLflow experiment tracking, DuckDB experiment catalog, and a contract-validated extension runner (7 extensions, 4 Tier-1 stable) — the model-management, observability, and automated-retraining pattern this role names, expressed under provenance-tracked governance.
- ▶ Shipped a deployment pathway suitable for governance-heavy environments: FastAPI inference (tabular / fusion / batch), RBAC with HMAC-signed tokens, hash-chained audit log at 21 CFR Part 11 posture, lab-hosted via Terraform + Helm. Runs on AWS / GCP. The biomedical modality set (ChEMBL, Hetionet, Open Targets, TCGA, GTEx, MIMIC-IV adjacent, eICU, BC5CDR) gives credible weight to PII/PHI-adjacent handling.
- ▶ Built the `ExperimentEngine` autonomous-evaluation loop: Gaussian-process surrogates with Expected-Improvement + Value-of-Information acquisition, sequential-probability-ratio-test stopping ($\alpha = \beta = 0.05$), and a DuckDB catalog. Validated at 1,037 experiments per hour with 104 discoveries per 1,100-experiment campaign (surrogate ECE = 0.055) — agent-evaluation guardrails at scale, first-party.
- ▶ Previously founded FIVSS: sole engineer for the founding stretch, running production rate-source API ingestion pipelines, warehousing millions of price observations in SQL, training gradient-boosted and time-series demand models, and shipping the full transactional web application. Acquired by Booking Holdings in 2022 — the production Python + SQL data-engineering credibility this role expects.
- ▶ Concurrently completing the Harvard ALM in Data Science & Artificial Intelligence (4.00 GPA), routing graduate final projects into SNPTX — DeepDDS graph-attention reproduction (CSCI E-104), eICU ICU-mortality forecasting under VAR / dynamic-factor / state-space models (CSCI E-116), ChEMBL32 graph analytics with NetworkX and Neo4j (CSCI E-108, Elston — assigned capstone advisor), BC5CDR biomedical NER (CSCI E-89b). Prior MITx MicroMasters in Statistics & Data Science.

Snakemake

DVC

MLflow

PyTorch

Graph Neural Nets

Diffusion Models

LangChain / LLM Agents

Python

Docker

CI / Reproducibility Gates

Content-Hashed Data

EVIDENCE

[SNPTX-ACADEMIC](#)

[REPRO-CHEMBL](#)

[REPRO-DEEPDDS](#)

[GITHUB](#)

FIVSS · Founder

IP ACQUIRED · BOOKING HOLDINGS

2012 – 2022 · ACQUIRED

FIVSS · predictive-pricing platform for five-star hotel-suite inventory · [acquired by Booking Holdings](#) (NASDAQ: BKNG; parent of Booking.com, Priceline, Kayak, OpenTable)

Founded and led FIVSS through a ten-year build, from an original insight — that the secondary market for luxury suite inventory was systematically mispriced — to a proprietary ML pricing engine, a live transactional marketplace, and ultimately a strategic acquisition by [Booking Holdings](#). Operated as sole founder for the majority of the company's lifespan, owning every function from model design to legal-entity formation to buyer negotiation.

- ▶ **Product thesis** Identified persistent inefficiencies in five-star suite pricing across metropolitan and resort markets. Built proprietary demand-forecasting and price-discovery models that flagged underpriced inventory in near-real-time for secondary-market resale.

- ▶ **Data & ML:** Designed and ran ingestion pipelines against rate-source APIs and structured scrapers; normalized and warehoused millions of price observations in SQL. Trained gradient-boosted and time-series demand models that generated per-suite fair-value estimates and rank-ordered opportunities by expected margin.
- ▶ **Platform:** Built the full transactional web application — inventory dashboard, buyer-facing UI, payments integration, and back-office reconciliation — as sole engineer for the founding stretch.
- ▶ **Commercial operations:** Formed the operating entity, negotiated supplier and partner agreements, ran the day-to-day P&L, and grew the business from bootstrapped prototype to a cash-flow-generating operator with recurring inventory relationships.
- ▶ **Exit:** Led the buyer-side workstream through diligence and close, culminating in the 2022 acquisition by Booking Holdings.

Predictive Pricing

Demand Forecasting

Gradient Boosting

Time-Series Models

SQL Warehousing

Python / ETL

Web Platform

Payments

Marketplace Ops

M&A Negotiation

Investment Sales & Trading · Vice President

PRE-2012

VP-level sales-and-trading roles in banking (Morgan Stanley, Merrill Lynch, etc.) including client coverage and market-facing execution in a large-account, high-throughput trading environment — the quantitative, commercial, and pressure-context substrate underneath the subsequent founder chapters.

SNPTX in brief

Public inspection surface for the current build. Every cell links to an artifact designed to be reviewed end-to-end without follow-up.

Deterministic harness

Seed pinned at [seed=42](#) across Python, NumPy, PyTorch, and CUDA. Environment locked via [pyproject.toml](#) plus lock file.

[METHODOLOGY](#) • [REPRODUCIBILITY](#) →

Five-minute reproduction

Carved-out repository reproduces the ChEMBL bioactivity GCN baseline end to end on commodity CPU in roughly five minutes, with stated headline metric.

[SNPTX-REPRO-CHEMBL](#) →

Inspection orientation

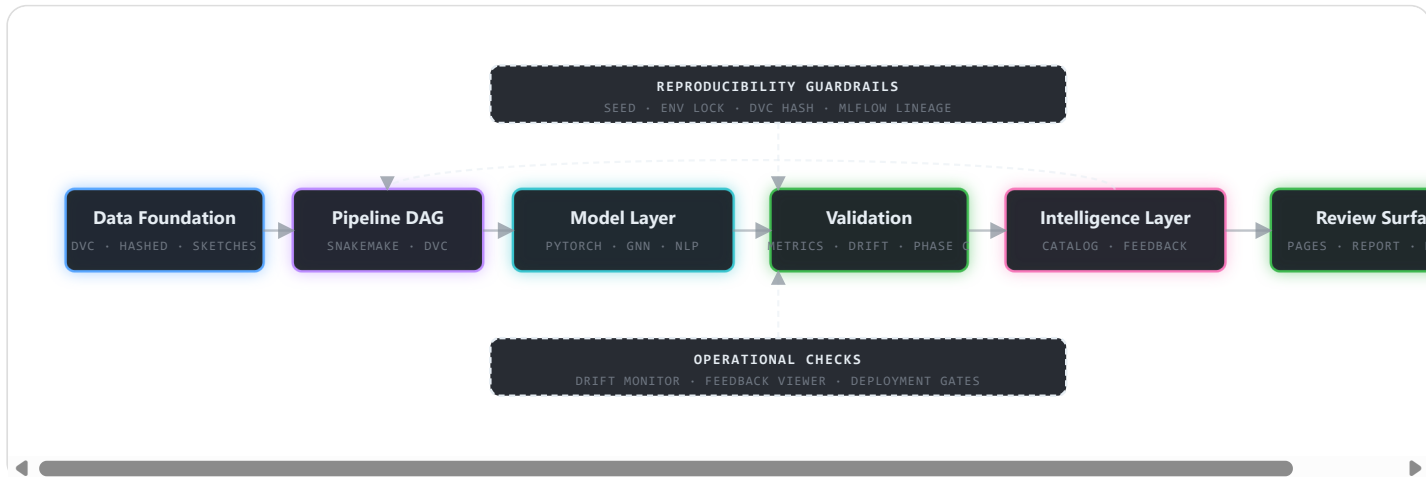
Short orientation page intended for senior technical readers, with three reading paths and an inspection-invariants panel.

[SNPTX-ACADEMIC](#) →

Methodology surface

Architecture, methodology, and references pages organize the design, the reproducibility controls, and the literature the work draws on.

[ARCHITECTURE](#) →



Selected technical work

Concrete, inspectable artifacts drawn from the SNPTX build and the concurrent Harvard graduate program. Each is designed to be reviewed end-to-end.

Five-minute reproduction repo

End-to-end reproduction of a ChEMBL bioactivity GCN baseline from a clean clone on commodity CPU. Stated headline metric, pinned environment, deterministic seed.

[reproduces in ~5 min • CPU-only](#)

[SNPTX-REPRO-CHEMBL](#) →

DeepDDS drug-synergy reproduction

Graph-attention drug-synergy model reproduced under the SNPTX deterministic harness. Full provenance and environment lock; served as the CSCI E-104 final project.

[A / 1.0000 raw score](#)

[SNPTX-REPRO-DEEPDDS](#) →

ICU mortality forecasting (eICU)

Temporal early-window ICU mortality forecasting on the eICU Collaborative Research Database, packaged as a self-contained reproducible notebook with dual real-data and synthetic-fallback execution paths.

[A • CSCI E-116 final](#)

[NOTEBOOK](#) →

ChEMBL32 bioactivity mining

Graph-analytic exploration of medicinal-chemistry data across ChEMBL32; scaled

similarity, clustering, and recommender workflows applied to real bioactivity records.

[A · CSCI E-108 final](#)

[NOTEBOOK →](#)

Education

Full academic detail — instructor-level coursework matrix, per-course raw scores, and named faculty references — available in the [detailed academic transcript](#).

ALM, Data Science & Artificial Intelligence

Harvard University (HES '27)

GPA 4.00 / 4.00 across all completed coursework. Capstone: deterministic reproducibility and provenance harness for multi-modal biomedical ML, advised by Stephen Elston, PhD.

MicroMasters, Statistics & Data Science

MIT Institute for Data, Systems & Society (IDSS) · MITx · 2022–2024

Full program awarded 2024. Rigorous probability, mathematical statistics, machine learning, and applied data-analysis sequence including the proctored capstone exam.

BS, Business Administration · Finance & International Business

Northeastern University

Undergraduate foundation in business administration, finance, and quantitative business analysis that complements the later founder/operator chapters.

GRADUATE COURSEWORK ([HARVARD ALM](#) · [MITX MICROMASTERS](#))

Advanced Deep Learning

Dynamic Modeling & Forecasting

Data Mining, Discovery & Exploration

Natural Language Processing

Epidemiology & Biostatistics

Foundations of Data Science & Engineering

Robotics, Autonomous Vehicles & AI

Machine Learning (MITx 6.86x)

Data Analysis: Modeling & Computation (6.419x)

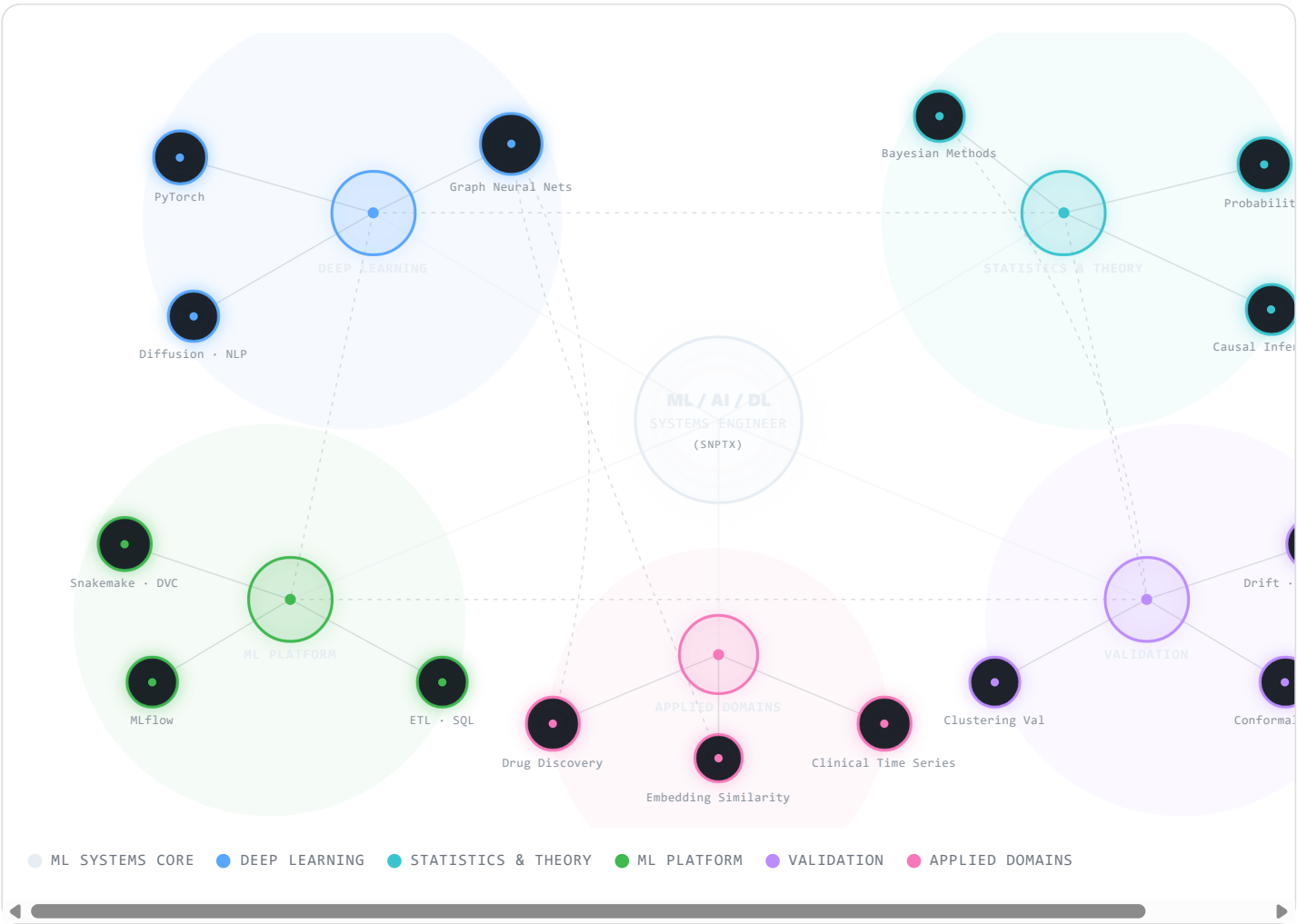
Probability (6.431x)

Fundamentals of Statistics (18.6501x)

MicroMasters Capstone Exam

Capabilities

The technical capabilities behind SNPTX, organized by dimension — deep learning, statistics and theory, ML platform, validation, and applied domains. The connections show where graduate coursework and hands-on system-building reinforce one another.



DEEP LEARNING & MODELING

- PyTorch
- Graph Neural Networks
- Transformers & NLP
- Diffusion Models
- Time-Series & State-Space Models
- Gradient Boosting
- Bayesian Inference
- Recommender Systems
- LLM Agents · LangChain

STATISTICS & THEORY

- Probability
- Mathematical Statistics
- Bayesian Optimization & Surrogates
- Causal Inference
- Experiment Design
- Hypothesis Testing

ML PLATFORM & DATA ENGINEERING

- Snakemake
- DVC
- MLflow
- Content-Hashed Pipelines
- Deterministic Runtimes
- Docker
- CI / Reproducibility Gates
- SQL (MySQL / Postgres)
- ETL & Ingestion Pipelines

VALIDATION & RELIABILITY

- Drift Detection
- Model Calibration
- Conformal Prediction
- Clustering Validation
- Coreset Sketches
- Provenance & Lineage
- Reproducible Research Infrastructure

APPLIED DOMAINS

Biomedical ML · Drug Discovery

Clinical Time Series

Predictive Pricing & Marketplaces

Embedding Similarity Search

● COMMERCIAL & OPERATING

Founder / Sole Operator

M&A · Buyer-Side Diligence

Entity Formation

P&L Ownership

Technical Writing & Reviewer Communication

References

Academic references drawn from graduate courses in which the final project fed directly into SNPTX. Each reference's scope is bounded to that course and its artifact.

Zoran B. Djordjević, PhD

REFERENCE

Harvard University · CSCI E-104 Advanced Deep Learning · Spring 2026 · A / 1.0000

Advanced deep-learning execution, graph-attention drug-synergy reproduction, and extension of course-level work into a deterministic SNPTX benchmark surface. Contact: zdjordj@fas.harvard.edu.

Stephen Elston, PhD

REFERENCE · CAPSTONE ADVISOR

Harvard University · CSCI E-108 Data Mining · Fall 2025 · A · assigned ALM capstone advisor

ChEMBL32 bioactivity mining and graph analytics, continuation into SNPTX seed datasets, and capstone supervision context. Contact: stephen_elston@g.harvard.edu.

Dmitry V. Kurochkin, PhD

REFERENCE

Harvard University · CSCI E-89b NLP · Fall 2025 · A

Biomedical NLP execution and hybrid statistical / neural NER over the BC5CDR corpus — the upstream text-processing components that now feed SNPTX. Contact: dkurochkin@fas.harvard.edu.